Marcos Galván López

• Mexico City, Mexico • magalop221@gmail.com • github.com/Magalop-bit •

Professional Profile

AI Engineer with a rare combination of applied mathematics, probabilistic modeling, and hands-on generative AI systems engineering. Passionate about pushing AI toward edge applications, spanning autonomous multi-agent research systems and gradient-free optimization of model weights. Research background in mechanistic interpretability of LLM weight space and fine-tuning techniques, complements hands-on experience architecting and shipping full-stack agentic systems (LangGraph orchestration, multi-model routing, sandboxed execution, persistent state, and production-ready desktop software). Comfortable owning an AI product end-to-end – from the mathematics behind an optimization method to the API layer and the interface that exposes it – and driven to stay current with the fastest-moving areas of applied AI.

Technical Skills

Programming Languages: Python, JavaScript, C++, C#, Java
Cloud & Infrastructure: AWS, Git/GitHub, Supabase/Postgres, relational databases
Agent & GenAI Engineering: LangGraph agent orchestration, Anthropic Claude SDK and Gemini API integration, multi-model cost-optimization strategies (free-tier vs. on-demand routing), LLM evaluation pipelines, LoRA and zero-order fine-tuning techniques, mechanistic interpretability of weight space, agent–data-structure design
ML & Math Foundations: Monte Carlo methods (MCMC), stochastic calculus
Other: Technical writing and documentation, cross-institutional collaboration, English C1 (TOEFL-iBT), continuous tracking of emerging AI research and tooling

Professional Experience

Banco de México — Mexico City, MX
Informatics Technician — July 2023 – January 2024

- Monitored and maintained database processes supporting critical financial services infrastructure, ensuring data integrity and availability for downstream systems.
- Designed and implemented secure file-transfer protocols between external clients and the central bank's internal systems.

Selected Projects

Auto Research Agent – Autonomous multi-agent system for scientific hypothesis exploration *GitHub*

- Architected a LangGraph-orchestrated pipeline that decomposes open-ended research questions into testable hypotheses, generates and executes feasibility code in an isolated sandbox, and scores each attempt against a fixed rubric (empirical evidence, theoretical soundness, generalization, model confidence).
- Designed a cost-aware multi-model strategy, routing high-volume idea generation to a free-tier model (Gemini Flash) and reserving judgment-critical steps – hypothesis decomposition, evaluation, synthesis – for Claude.
- Built the system end-to-end: FastAPI backend, Supabase/Postgres persistence layer for the full exploration log, and a React + Tailwind + shadcn/ui frontend for real-time visualization of agent attempts.

- Positioned the exploration log itself – not only the final output – as the core product, differentiating the tool's value proposition from comparable autonomous-research systems.

LoRA-RandOpt – LoRA-restricted random search for LLM specialization *GitHub*

- Built and documented an evaluation pipeline for Qwen2.5-1.5B-Instruct on GSM8K and BBH Boolean Expressions benchmarks under GPU-constrained hardware.
- Implemented zero-shot baselining, top-K majority-vote ensembling, and rigorous statistical comparison (Mann–Whitney U) between fine-tuning strategies.

Education

Centro de Investigación en Computación (CIC) Mexico City, MX
Ongoing – MSc in Computer Science. GPA 3.9 Graduation Date: 01/27

- Research in probabilistic AI for semantically accurate latent-space representation and its application to pattern classification and zero-shot learning.

University of Edinburgh (UoE) Edinburgh, UK
Academic Visit – Artificial Intelligence Applications Institute September 2025 – December 2025

Instituto Politécnico Nacional (IPN) Mexico City, MX
Mathematical Engineer. GPA 3.6 Graduation Date: 08/23

Publications & Talks

- Galván-López, M., Upreti, N., & Belle, V. *Soft Symbol Grounding for Prototypical Concepts* (Accepted). 20th International Conference on Neurosymbolic Learning and Reasoning (NeSy 2026), Lisbon, Portugal, April 2026.
- Galván-López, M., Calvo, H., & Valdez-Rodríguez, J. E. (2025). *Representation of Neuro-Symbolic Networks for Sub-algebraic Terms.* Mexican Congress on Artificial Intelligence, pp. 174–186. Springer Nature Switzerland. `https://doi.org/10.1007/978-3-031-97910-1_14`

Extra-Curricular & Academic Service

- **Judge & Speaker**, AI Safety Hackathon by Apart Research
- **Mentor**, Apart Research AI Safety Fellowship
- **Creator & Instructor**, "Mi Primer GPT" – introductory course on language generation with Transformers; first course of its kind delivered in Spanish
- Member, Mexican Association of Artificial Intelligence (SMIA)